Northlake University — Implementation Discovery Questionnaire (Completed)

DATE October 1, 2026
PREPARED BY Northlake University

Prepared by: Campus Operations, with Facilities, Finance, Housing, and Sustainability. Packet version: 0.4.

This is our response to your standard discovery questionnaire. Answers reflect how we operate today and what we want the new system to do.

1. Organization & goals

1.1 Describe your organization. Northlake University is a mid-size university operating a main academic campus, a student housing property (Cedar Row Apartments), a central thermal plant run as its own operating company (Northlake Thermal Plant), and — in a later phase — a mixed-use property (Northlake Town Center). Facilities Operations manages campus utilities, building automation, and meters. Housing Operations manages Cedar Row. The thermal plant has its own manager and operator, and since July 2024 it also owns Ridgeline Wind, a ten-turbine wind site with its own site manager, which doubles as the Mechanical Engineering department's Wind Energy Research Lab. Finance owns external utility payment and internal cost allocation. The Sustainability Office owns reporting.

1.2 What are your top goals for this system?

1. One source of truth for all utility usage and cost across properties.
2. Accurate monthly bill processing, validation, and payment/AP export.
3. Internal cost allocation for our central thermal plant.
4. Resident rebilling from our Cedar Row apartment submeters, and later tenant rebilling at Town Center.
5. Sustainability reporting: ENERGY STAR scores and greenhouse-gas emissions.
6. Scheduled reports delivered automatically to building and finance staff.
7. Wind performance: what each Ridgeline turbine made against what the wind made possible, and the energy the farm should have produced over any period.

1.3 What does success look like in 12 months? Every property's bills load and reconcile automatically; Finance exports approved costs to our ERP; tenants are billed accurately; we submit ENERGY STAR scores and report emissions by scope; and ~120 staff receive the reports they need without asking us to run them.

2. Scope

2.1 Properties in scope. Northlake Main Campus (seven buildings: Admin Hall, Science Center, Library, Student Center, Lakeview Residence Hall, the Recreation and Aquatics Center and the Parking Structure, plus Campus Grounds and onsite solar), Northlake Thermal Plant (the Central Plant, its own operating company, and the Ridgeline Wind Site it bought on July 1, 2024: a turbine field and a substation near Rio Vista), Cedar Row Apartments (Cedar Row A and B with 124 apartments, the Common House, and the carport and grounds). Northlake Town Center (retail + office tower + annex + infrastructure) is a later phase.

2.2 Commodities. Electricity, natural gas, water, sewer, stormwater, chilled water, steam, onsite solar generation and off-site wind generation.

2.3 In scope now vs later. Town Center tenant rebilling and EnergyAI insights are desired but we understand they may follow once the core is live.

3. Utilities & billing

3.1 Who are your providers? Valley Electric District (electric), Sierra Gas Utility (gas), River City Utilities (bundled water/sewer/stormwater), Northlake Thermal Plant (internal chilled water + steam), Helios Onsite Solar (PPA), and Ridgeline Wind (the plant's own wind company, sold to the plant at an internal transfer price).

3.2 How do bills arrive today? A mix of vendor billing files and PDF statements. Electric and water can be delivered as files; gas and the solar PPA are entered from statements. Ridgeline's monthly transfer statement is built by hand today; we want the system to generate it from the turbine meters. Tariff sheets and samples are enclosed.

3.3 Do you have time-of-use / demand / tiered rates? Yes — TOU and billing demand on electric, tiered water, seasonal gas, internal allocation rates, a PPA, a generator interconnection schedule whose export credit Valley Electric moves to the Central Plant's bill, and Ridgeline's internal transfer price. See the rate tariff sheets. We also want to model a real-time-pricing option and a large-demand ratchet rate for analysis.

3.4 How are utility costs coded for accounting? By GL account plus our five coding segments (Fund, Department, Program, Activity, Object). See the GL guide. We export approved costs to our ERP for payment.

4. Metering & data

4.1 What meter/interval data do you have? 15-minute electric interval at Science Center, Student Center, Lakeview Residence Hall, the Recreation and Aquatics Center and the Central Plant; 15-minute solar generation/export; hourly chilled-water and plant production; hourly electric and monthly water submeter reads for every Cedar Row apartment; monthly reads elsewhere. Water/gas include handheld route reads. At Ridgeline: 10-minute turbine SCADA (energy, power, the controller's possible power, nacelle wind, rotor speed and run state) for each of ten turbines, a 10-minute met mast in metric units, and Valley Electric's 15-minute interconnection meter in both directions with reactive energy.

4.2 What source systems feed data? Utility billing files, meter-data-service interval files, AcquiSuite and MV90 files, central-plant BACnet and Modbus points, wireless sensor files, fixed-network submeter files, a SQL historian, NOAA and Aeris weather, handheld route uploads, and Ridgeline's SCADA file export, whose clock stays on Pacific Standard Time all year. See the Source System Inventory.

4.3 Weather. We use public airport stations: KSAC (Sacramento Executive Airport) is the primary station for the campus and Central Plant buildings; Cedar Row benchmarks on KSMF (Sacramento International Airport),

which is also our forecast source; and the Ridgeline site uses KSUU (Travis Air Force Base), whose wind is reported in knots, as the long-term reference for its own met mast. We want heating and cooling degree days derived from temperature.

5. Users & access

5.1 How many users? ~23 core operational users across the three properties plus implementation/support, and a broader audience of ~120 building representatives who should receive scheduled reports.

5.2 How do you think about roles? Company administrators, facilities/utility setup, rates, bill entry, bill import, report viewers, sustainability reporters, auditors, plant operators, a thermal allocation accountant, wind operations limited to the Ridgeline site, and university researchers who may read the Ridgeline site and nothing else of the plant. See the User Access Request. We require approval authority separation (the person who enters a bill is not the person who approves it).

6. Reporting

6.1 What reports matter most? Cost and usage by property/building, bill variance, fiscal-period processing, rate analysis, tenant bills, ENERGY STAR ratings, greenhouse-gas reports, and wind generation against the energy the wind made possible, per turbine and for the farm.

6.2 Delivery. Monthly building energy report to building representatives; weekly cost rollup to Finance; daily operations report to operations staff. We'd like these emailed automatically.

7. Current pain points

- Utility data lives in spreadsheets and vendor portals; no single rollup.
- Manual bill entry is slow and error-prone; no validation.
- Thermal plant cost allocation is a manual monthly exercise.
- No tenant rebilling system for Town Center; done by hand.
- Sustainability reporting is assembled manually each year.
- Ridgeline's transfer statement and turbine performance are worked out by hand from SCADA exports, and the researchers ask us for files.

8. Known events & data-quality notes

We have documented intentional events (meter replacement, estimated → corrected bill, account-number change, rate change, a duplicate billing file, a missing interval block, the Ridgeline acquisition and its generator account transfer, the first bill credit transfer, and a turbine gearbox outage) in the overview and workbook so they can be handled deliberately.

9. Key contacts

See the cover letter contact table. Project sponsor: Samantha Ireland.